Service Plus: Proposed Architecture

All 25 modules on WSO2 Identity Server and API Manager, with an event backbone, one core database and a separate integration layer.

Service · Request · Event

M1 to M25 are the module numbers.

1 Channels

Citizen web portal
React, forms from definitions
M8

Mobile app
Same services and forms
M8

Operator console
Assisted applications
M13

Officer workspace
Work lists and decisions
M9

Admin and Service Studio
Design and publish services
M4

HTTPS

2 Edge and security

Reverse proxy and web firewall
TLS, security headers, rate limits. Routes portal and API traffic to the gateway.
M3

3 Gateway and identity (WSO2)

WSO2 API Manager
The one front door. Token and ApiKey checks, throttling, API versions, developer portal for departments.
M14

WSO2 Identity Server
Sign-in, OTP and MFA, roles from postings, tokens for citizens, officers, operators and departments.
M1

one front door to every service, token checked first

4a Identity and offices

Account service
Registration, verification, officer onboarding
M1

Office directory
Offices, areas, dated postings
M2

Access check
Role + office + area + case
M3

Operator service
Centres, operators, assisted by
M13

4b Service setup

Service catalogue
Service types and launch
M4

Form builder
Versioned forms, one renderer
M5

Rules engine
Papers and fees as tables
M6

Publishing
Draft, test, approve, go live
M17

Derived services
Follow-on service offers
M22

Service search
Search in the citizen's language
M25

4c Application work

Application service
Draft, submit once, track
M8

Workflow engine
Flowable (BPMN): steps, actions
M7

Task API
Work lists, claim, decide
M9

Bulk processing
Many cases, one request
M21

Time limits
Clocks and escalation
M24

Grievances and appeals
Up to three levels
M15

4d Documents and money

Document service
Certificates, signing, QR check
M10

Payment service
Fees, receipts, matching
M11

Notification service
SMS, email, WhatsApp
M12

Benefit payments
After approval, kept separate
M23

4e Platform and support

Reports
Dashboards and exports
M16

Audit and operations
Every action, monitoring, recovery
M18

Platform settings
Name, languages, gateways, calendar
M19

Migration and import
Existing services and applications
M20

Npgsql · functions

events

outside calls

5 Data and streaming

PostgreSQL 18
One core database
Function-only access

Redis 8
Sessions, token verdicts

Apache Kafka 4.3
Outbox events: submitted, paid, decided, issued

Object storage
Papers and issued certificates

OpenSearch
Service and case search

Reporting store
Read copy for dashboards

Event consumers
Notifications, payments matching, audit, reports and search all react to events, so a failure in one never undoes a decision.

Keys and secrets
A secrets vault and a signing-key module. Encrypted storage and backups, restores rehearsed.
M18

6 Integration layer

WSO2 Micro Integrator
The single connection point: retries, timeouts, logs and a named owner for every outside link.
M14

7 External systems

Identity: Aadhaar, DigiLocker

Signing: eSign

Payment gateways and treasury

PFMS and DBT

SMS, email, WhatsApp

Departments and their portals